Lab 3: Architecture Justification

Digital Campus Library Reservation Gateway | Team 08 | Krrish Singla | PES1UG24AM141 | Section C

Architecture Selection

We chose **Layered Architecture** for the **Digital Campus Library Reservation Gateway** System. The web portal is the Presentation layer, the Catalog Service, Hold Manager and Notification Service make up the Business layer, and the Library Database is the Data layer.

Architectural Style Comparison

Style	What works for the library gateway	What doesn't work	Fit
Layered	Easy to build and test. UI, hold rules and data access stay in separate layers, and a single database makes it easy to keep the hold queue in order.	To scale it you have to copy the whole business tier, and the layers can end up depending on each other too much.	High
Client-Server	One central server, so control is simple and deployment is the easiest of the three.	If that server goes down everything stops, and it becomes a bottleneck when many students search at exam time. UI code and library rules also get mixed together.	Low
Microservices	Every feature can be scaled and can fail on its own without taking the others down.	Needs a lot of network calls and operations work for a small team, and splitting the data makes it hard to keep the hold queue consistent.	Medium

Two Reasons

- **The hold queue has to stay fair and in order.** Our requirements (FR-001, FR-002) say holds follow first-in-first-out, a member can have at most 5 holds, and a held copy is released after 48 hours. All of that has to be checked in one place using one set of data. In the layered design the Hold Manager does this using a single Library Database. With microservices the hold, catalog and notification data would live in different services, and keeping the queue correct would be harder.
- **It fits the size of the project.** This is one campus library with about 50,000 titles and a small team, so running many separate services would be more work than it is worth. Each component talks through an interface (ICatalog, IHold, INotify and the data interfaces), so we can change the email/SMS provider or the database later without touching the UI. We can also run the business layer on two nodes to help reach the 99.9% availability goal (NFR-002).

Security Advantage

The web portal never connects to the database directly. Every request goes through the Business layer first, where we check that the member is logged in and allowed to do the action. A Student Member can place holds, but only the Head Librarian can change the catalog or the hold queue. The database only accepts connections from the Business layer, so there is one place to enforce security and fewer ways to attack the system.

Performance Benefit

A catalog search is a direct call from the Catalog Service to an indexed, paginated query, with no network hops between services. This helps us reach the 150 ms goal for 50,000 titles (NFR-001). When a held book is returned, the alert is sent to the Notification Service in the background, so placing a hold stays fast and the member is still told within the 5-minute limit.

Component Diagram

